

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code :

```
#include <stdio.h>

int main()
{
    int frames[10], pages[50];
    int n, f, i, j, k, flag, pos, faults = 0, hits = 0;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i=0; i<n; i++)
        scanf("%d", &pages[i]);

    printf("Enter number of frames: ");
    scanf("%d", &f);

    for(i=0; i<f; i++)
        frames[i] = -1;

    for(i=0; i<n; i++)
    {
        flag = 0;

        for(j=0; j<f; j++)
        {
            if(frames[j] == pages[i])
            {
                hits++;
                flag = 1;
                break;
            }
        }
    }
}
```

```
if(flag == 0)
{
    int farthest = -1, index = -1;

    for(j=0;j<f;j++)
    {
        int found = 0;

        for(k=i+1;k<n;k++)
        {
            if(frames[j] == pages[k])
            {
                if(k > farthest)
                {
                    farthest = k;
                    index = j;
                }
                found = 1;
                break;
            }
        }

        if(found == 0)
        {
            index = j;
            break;
        }
    }

    frames[index] = pages[i];
    faults++;
}

printf("\nTotal Page Faults = %d", faults);
printf("\nTotal Page Hits = %d", hits);

printf("\nPage Fault Percentage = %.2f%%", (float)faults/n*100);
printf("\nPage Hit Percentage = %.2f%%", (float)hits/n*100);

return 0;
```

Output:

```
Enter page reference string:
```

```
1 2 3 4 1 2 5 1 2 3 4 5
```

```
Enter number of frames: 3
```

```
Total Page Faults = 7
```

```
Total Page Hits = 5
```

```
Page Fault Percentage = 58.33%
```

```
Page Hit Percentage = 41.67%
```